

Machine Learning and its Applications

Today's goals

Module 1

Machine Learning Fundamentals I: Basic Concepts

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 8 Introduction

Introduction to Module 8's Study

Video • 1 min

What Does “Good” Mean in Machine Learning Models?

Reading • 5 min

Module 8-1 ML Model Training Fundamentals

ML Model Training Fundamentals: Parameters, Hyperparameters, and Loss Functions

Video • 8 min

ML Model Training Fundamentals: Loss Functions and Gradient Descent

Video • 5 min

ML Model Training Fundamentals: Gradient Descent

Video • 10 min

Materials for ML model training

Reading • 10 min

What Does It Actually Mean to “Train” a Machine Learning Model?

Reading • 5 min

Module 8-2 ML Model Evaluation Fundamentals

ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation

Video • 13 min

ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation

Video • 5 min

Materials on ML Model Evaluation Fundamentals

Reading • 10 min

When a Model Fails: Is It Bias, Variance, or Something Else?

Reading • 5 min

Module 8-3 Summary of the ML Process

Summary of the ML Process

Video • 5 min

Materials for ML Process

Reading • 10 min

From Data to Decisions: Reconstructing the Machine Learning Pipeline

Reading • 5 min

Module 8 Learning Coach Dialogue

• 5 min

Module 8 Quiz

Module 8 Quiz

Practice Assignment • 30 min

Module 8 Assignments

Module 8 Assignment 1

Graded App Item • 30 min

Module 8 Assignment 2

Graded App Item • 30 min

CSV file for Module 8 Assignment

Reading • 1 min

Module 8 Recap

Reading • 5 min

Module 3

Data Preparation

Module 4

Support Vector Machines

Module 5

Artificial Neural Networks

What Does It Actually Mean to “Train” a Machine Learning Model?

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In this part of the module, training a machine learning model was reframed as an **optimization problem**: choosing model parameters that minimize a loss function.

Reflect on the following points:

1. What is being learned—and what is not

- In the models discussed so far, what quantities are **learned from data**, and what quantities must be **chosen in advance**?
- Why is it important not to confuse **model parameters** with **hyperparameters**?

2. Loss functions as design choices

- Why does training require a loss function at all?
- For regression problems, what trade-offs exist between using MSE and MAE?
- How can a poorly chosen loss function lead to misleading training outcomes, even if optimization succeeds?

3. Training as optimization

- Why can model training be viewed as minimizing a function rather than “learning rules”?
- What role does the **gradient** play in this process?
- Why is the learning rate critical, and what could go wrong if it is chosen poorly?

4. Efficiency and scale

Compare batch, stochastic, and mini-batch gradient descent:

- What problem does each variant try to solve?
- Why is mini-batch gradient descent the default choice in practice?

In what situations might a simpler method still be preferable?

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https://www.coursera.org/learn/machine-learning-and-its-engineering-applications/supplement/YTYbA/what-does-it-actually-mean-to-train-a-machine-learning-model

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